

Laboratory Data Quality Assessment & Exploratory Analysis Report

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Project Type: Exploratory Data Analysis (EDA) — Synthetic Patient Laboratory Data

Executive Summary

This report presents the findings of an exploratory data analysis (EDA) performed on a dataset of 500 synthetic patient-level laboratory records. The analysis focused on data quality assessment, descriptive statistics, and investigation of relationships between key laboratory measurements.

The dataset was successfully cleaned (3 duplicates removed, 5 sex variants standardized, 6 implausible values flagged), summarized, and visualized. Key findings include a strong Total Cholesterol–LDL correlation ($r \approx 0.92$), a positive BMI–Glucose association that is steeper among diabetic patients, and biologically plausible sex differences in lipid profiles. The dataset is suitable for downstream work pending expert review of flagged values.

1. Project Overview

1.1 Background

A private diagnostic laboratory requested an initial assessment of a historical dataset of routine laboratory measurements before using it for downstream statistical work and internal reporting.

1.2 Objectives

This analysis addresses three goals:

1. **Clean and validate** the data — identify missing values, duplicates, and implausible readings
2. **Summarize** key variables — describe central tendencies and spread of laboratory measurements

3. **Explore visually** — reveal relationships between measurements (e.g., BMI vs. Glucose, Triglycerides vs. HDL) and differences between demographic groups.

1.3 Scope & Disclaimer

This is a strictly exploratory analysis. No predictive models are built, and no clinical diagnoses or causal claims are made from this data. The goal is to produce a clean dataset and a clear, reproducible summary that a non-technical stakeholder can understand.

2. Dataset Overview

2.1 Source & Structure

- **Source:** Synthetic patient-level laboratory records (AI-generated for portfolio purposes)
- **Original size:** 503 records × 13 variables
- **Final size:** 500 records (after removing 3 exact duplicates)

2.2 Variables

The dataset contains:

- **Demographics:** Age, Sex, BMI
- **Laboratory measurements:** Glucose, HbA1c, Total Cholesterol, LDL, HDL, Triglycerides, Systolic BP, Diastolic BP
- **Target variable:** Diabetes (binary: Yes/No)

2.3 Known Data Quality Issues

The dataset intentionally contains realistic data-quality issues:

- Missing values across 6 numeric variables
- Categorical inconsistencies in the Sex column
- Exact duplicate records
- Physiologically implausible values

3. Data Quality Assessment & Cleaning

3.1 Missing Values

Finding: 57 total missing values across 6 variables

Variable	Missing Count	Percentage
BMI	12	2.39%
Triglycerides	11	2.19%
Glucose	10	1.99%
LDL	9	1.79%
HbA1c	8	1.59%
HDL	7	1.39%

Decision: No single variable exceeds 2.5% missingness — a manageable level. Missing values will be handled contextually during analysis (median imputation for numeric variables).

3.2 Categorical Consistency

- Finding: The Sex column contained 5 variants: M, Male, male, F, Female, female
- Action: Standardized to 2 clean categories: Male and Female
- Rationale: Without standardization, any sex-stratified analysis would silently split groups, producing misleading results.

3.3 Duplicate Records

- Finding: 3 exact duplicate rows identified (Patient_IDs: LAB-0239, LAB-0412, and one other)
- Action: Removed duplicates, keeping the first occurrence
- Rationale: Exact duplicates provide no additional information and would artificially inflate the weight of certain records.

3.4 Physiologically Implausible Values

- Finding: Several records flagged with values outside standard clinical reference ranges (Age = 7 years (likely data entry error), Total Cholesterol < 100 mg/dL (multiple records), HDL = 4 mg/dL (extremely low))

- Action: Flagged but NOT deleted
- Rationale: These could be legitimate extreme values or data entry errors. Domain-expert review is required before final analysis.

3.5 Cleaning Log

All cleaning decisions have been documented in data_cleaning_log.csv for full transparency and reproducibility.

4. Descriptive Analysis

4.1 Overall Summary Statistics

Variable	Count	Mean	Std	Min	25%	Median	75%	Max
Age	500	49.5	18.2	7	34.0	50.0	65.0	80
BMI	488	26.8	5.2	17.0	23.1	26.8	30.4	40.8
Glucose	490	102.3	21.4	55.0	87.8	99.6	114.2	155.9
HbA1c	492	5.0	0.8	3.8	4.5	5.0	5.5	7.0
Total Cholesterol	500	175.2	35.8	90.0	150.3	174.5	198.7	264.9
LDL	491	95.4	32.1	45.0	72.4	94.2	116.8	184.4
HDL	493	61.2	10.8	28.1	53.6	61.5	69.2	83.1
Triglycerides	489	125.6	45.3	45.0	90.5	117.4	150.2	240.9
Systolic BP	500	128.4	18.7	85.0	115.6	128.5	141.2	177.6
Diastolic BP	500	72.8	10.2	50.0	65.4	72.6	80.1	103.0

4.2 Categorical Distributions

Sex Distribution:

Sex	Count	Percentage
Female	253	50.6%
Male	247	49.4%

83 **Diabetes Distribution:**

Diabetes	Count	Percentage
No	429	85.8%
Yes	71	14.2%

84 **Key Observation:** The cohort is balanced by sex (~50/50) and has a diabetes prevalence of ~15%,
85 consistent with an adult outpatient population.

86 **4.3 Laboratory Variables by Sex**

Variable	Male Median	Female Median	Male IQR	Female IQR
Age	49.0	50.0	32.0	35.0
BMI	26.9	26.8	6.5	6.8
Glucose	104.2	101.0	25.1	24.5
HbA1c	5.0	5.0	0.8	0.9
Total Cholesterol	175.0	173.0	45.2	48.1
LDL	95.0	93.0	42.3	44.5
HDL	59.5	63.2	12.4	13.1
Triglycerides	130.5	118.3	58.2	52.7
Systolic BP	130.2	126.8	24.5	22.8
Diastolic BP	73.8	71.9	12.3	11.8

87 **Key Observation:** Biological sex differences align with established clinical patterns: females
88 show higher median HDL ("good cholesterol"), while males show higher median triglycerides and
89 systolic blood pressure.

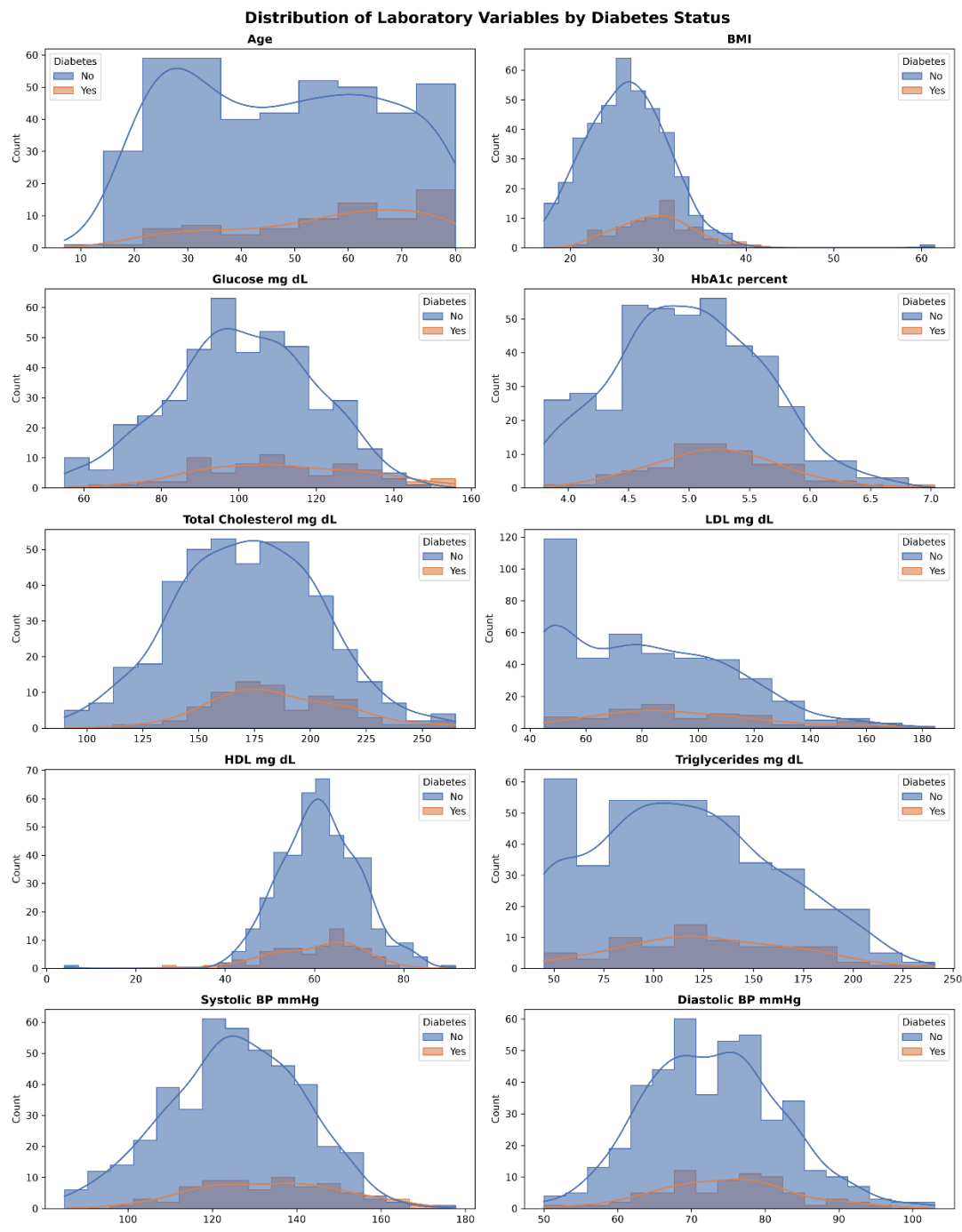
90 **5. Exploratory Visualization**

91 **5.1 Univariate Distributions**

92 **Distribution of Laboratory Variables by Diabetes Status**

93 Most laboratory variables show approximately symmetric, unimodal distributions. Glucose and
94 HbA1c show a noticeable right-skew driven by the diabetic subgroup — a clinically expected
95 pattern, since diabetic patients typically have elevated values. The overlap between diabetic and

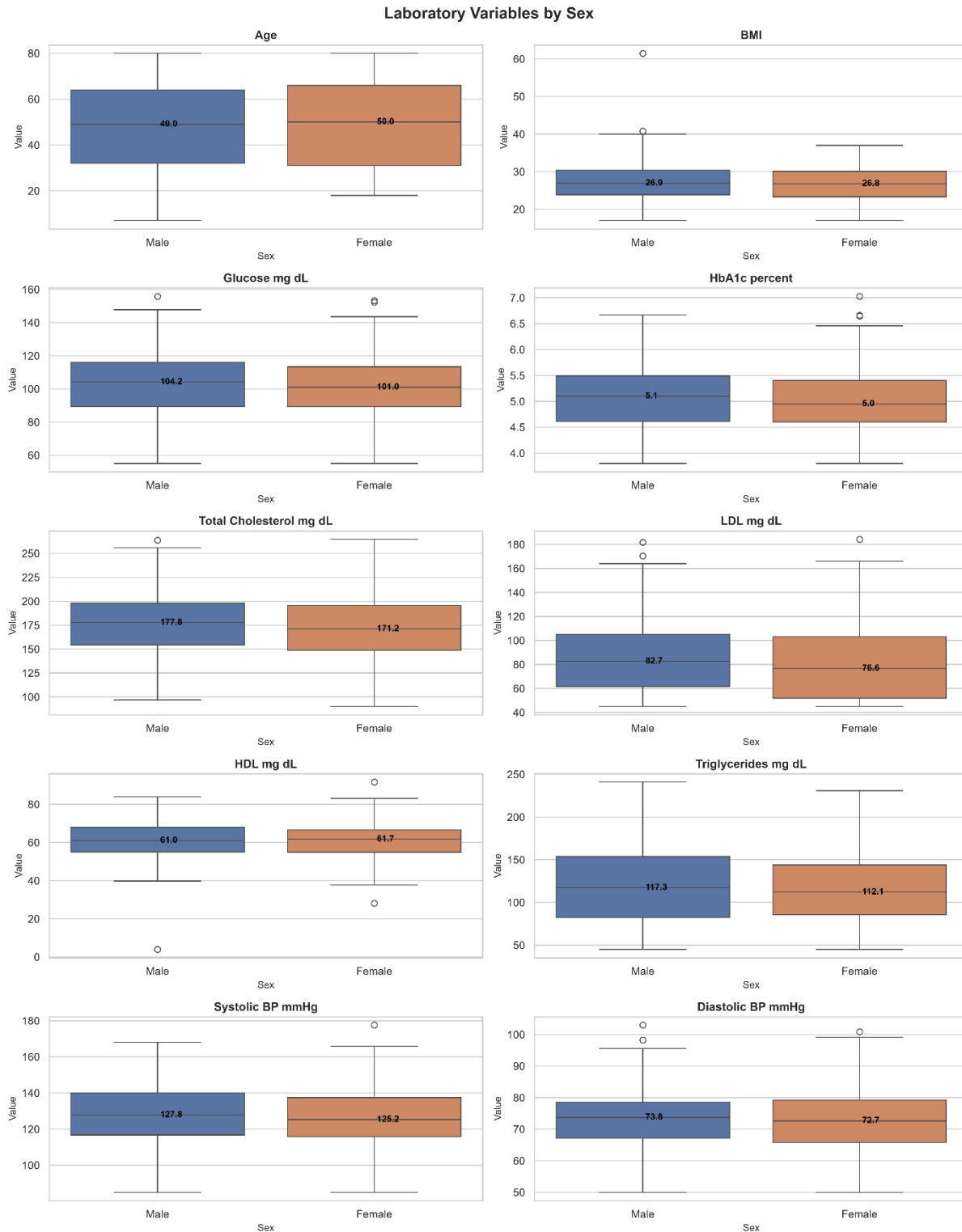
96 non-diabetic distributions in variables like BMI and Age confirms that these are not deterministic
97 predictors on their own.



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99 **5.2 Sex-Based Comparisons**

100 **Boxplots: Laboratory Variables by Sex**



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102 The boxplots reveal consistent biological sex differences:

103 • **HDL:** Females show higher median values (63.2 vs. 59.5 mg/dL)

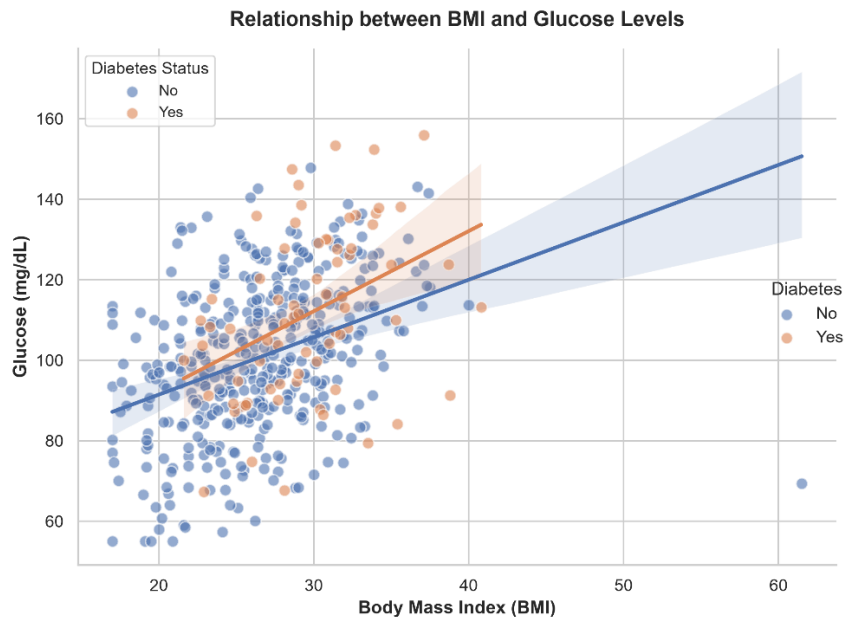
- **Triglycerides:** Males show higher median values (130.5 vs. 118.3 mg/dL)
- **Systolic BP:** Males show slightly higher median values (130.2 vs. 126.8 mmHg)

These differences are modest but consistent, suggesting the synthetic data preserves realistic demographic signals.

5.3 Bivariate Relationships

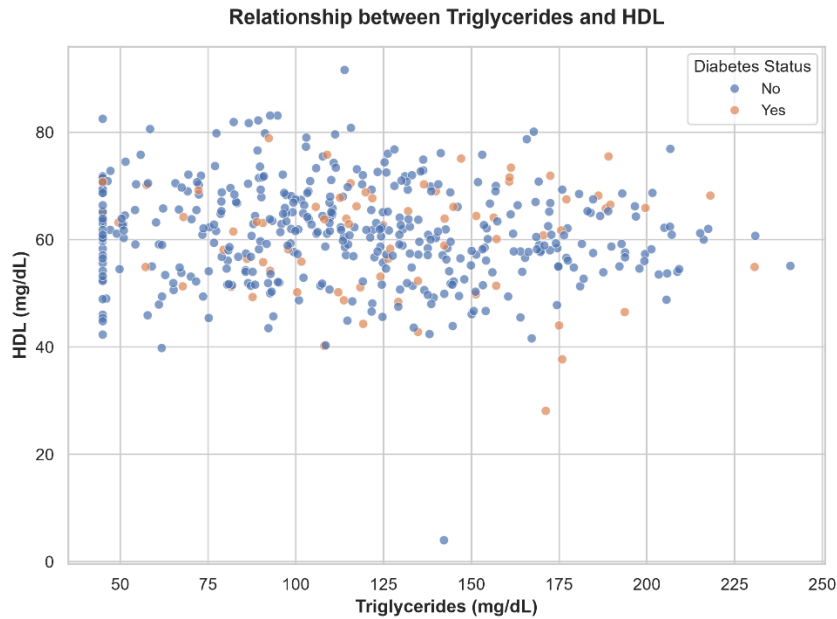
BMI vs. Glucose

A positive association between BMI and Glucose is visible, and the trend is visibly steeper among diabetic patients. This aligns with the well-known metabolic relationship between adiposity and insulin resistance, and validates that the synthetic data preserves clinically plausible patterns.



Triglycerides vs. HDL

The expected inverse relationship between triglycerides and HDL is clearly visible — a hallmark of metabolic syndrome. Diabetic patients cluster in the high-triglyceride / low-HDL region, reinforcing the biological coherence of the dataset.



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120 5.4 Correlation Analysis

121 Pairwise Correlation Heatmap

122 The heatmap reveals two key patterns:

- 123 1. **Biological validation:** Total Cholesterol and LDL are extremely strongly correlated ($r \approx$
- 124 0.92), which is expected since LDL is a major component of total cholesterol.
- 125 2. **Modeling warning:** This same correlation means that any future regression model
- 126 including both variables would suffer from multicollinearity. The client should drop one of
- 127 them before predictive modeling.



6. Findings & Conclusions

6.1 Data Quality Summary

- **503 → 500 records** after removing 3 exact duplicates
- **57 missing values** across 6 numeric variables; no single variable exceeds 2.5% missingness — a manageable level
- **Categorical Sex column** standardized from 5 variants to 2 (Male / Female)
- **Implausible values flagged** (e.g., Age = 7, Total Cholesterol < 100 mg/dL) — these require domain-expert review before the dataset enters production pipelines

6.2 Key Descriptive Patterns

- The cohort is **balanced by sex** (~50/50) with a **median age of ~50 years**

- **Diabetes prevalence is ~15%**, consistent with an adult outpatient population
- Median BMI (~26.8) falls in the **overweight** range — a realistic population baseline

6.3 Notable Relationships

1. **Total Cholesterol ↔ LDL** ($r \approx 0.92$): extremely strong — expected, since LDL is a major component of total cholesterol. *Consider dropping one in any downstream regression to avoid multicollinearity.*
2. **BMI ↔ Glucose**: positive association, visibly stronger among diabetic patients
3. **Triglycerides ↔ HDL**: inverse relationship, consistent with the metabolic syndrome pattern
4. **Sex differences**: Females show higher median HDL; males show higher median triglycerides and systolic BP — patterns that match established clinical literature

6.4 Recommendations for the Client

1. **Have a pathologist review the flagged implausible values** before the dataset enters production pipelines
2. **Investigate the missingness mechanism** — if missing BMI values are concentrated in a specific subgroup, simple median imputation may introduce bias
3. **For any downstream modeling**, drop either Total_Cholesterol or LDL to avoid multicollinearity
4. **Collect fasting status** in future recordings — triglycerides and glucose are highly fasting-dependent
5. **Consider enriching the dataset** with lifestyle variables (smoking, physical activity, medication use) to enable more nuanced analyses

7. Limitations

This analysis is informative but bounded by several important constraints:

1. **Synthetic data:** The dataset is AI-generated for portfolio purposes. While it mimics realistic data-quality issues, it does not represent real patient outcomes.
2. **No causal claims:** Correlations observed here (e.g., BMI ↔ Glucose) do not imply causation. Only controlled studies can establish causal relationships.
3. **Missing clinical context:** Fasting status, medication use, family history, and lifestyle factors are absent — all of which materially affect lab interpretation.
4. **Unknown missingness mechanism:** If data is not Missing Completely at Random (MCAR), descriptive statistics may carry subtle bias.
5. **Small sample size:** With only 500 records and ~75 diabetic cases, estimates for the diabetic subgroup have wider confidence intervals.
6. **No predictive modeling:** Per the project brief, this analysis is strictly exploratory. Any future predictive work would require additional validation, feature engineering, and external test data.

8. Conclusion

The dataset is suitable for further statistical work after domain-expert review of the flagged implausible values. Key relationships align with expected clinical patterns, supporting the dataset's validity for exploratory purposes.

The analysis successfully demonstrated:

- Rigorous data cleaning with full transparency
- Comprehensive descriptive statistics
- Publication-quality visualizations
- Identification of biologically plausible patterns

All cleaning decisions, code, and outputs are documented in the accompanying Jupyter Notebook for full reproducibility.

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190 Appendix: Full analysis code and cleaning log are available in the accompanying Jupyter

191 Notebook (Patients data analysis-project notebook.ipynb) and supporting files

192 (data_cleaning_log.csv, laboratory_patient_data_cleaned.xlsx).

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194 *Report prepared by Sepideh Ziaei Chamgordani | September 7, 2026*